**23** A student is investigating two electromagnetic waves, X and Y, in a vacuum.

Wave X has a wavelength of $5.2 \times 10^{-7}$ m. Wave Y has a frequency of 9.4 GHz.

Which principal regions of the electromagnetic spectrum contain waves X and Y?

| | X | Y |
|---|---|---|
| **A** | radio wave | ultraviolet |
| **B** | ultraviolet | visible |
| **C** | visible | microwave |
| **D** | microwave | radio wave |